Navaneethan Jeyaprakash

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Profile Summary

I have about 5 years and 7 months of IT experience, with solid skills in software development and data engineering. This gives me a well-rounded ability to build apps and handle big data.

I have more than 3 years in web app development, 2 years in Generative AI, and 1 year in data engineering. This includes strong skills in designing, testing, and maintaining software, plus handling data changes with tools like PySpark and SQL on Databricks.

In my last role as a Senior Data Engineer, I focused on creating and improving data pipelines to make data processing faster and better. I work with huge datasets, do tricky data changes, and use cloud tools.

Now, I'm looking to switch to a Data Science job focused on Generative AI. I can use my data engineering and coding skills to create new AI ideas. In my 2 years of hands-on GenAI work, I've learned key skills like fine-tuning large language models (LLMs), crafting prompts, and building AI agents with tools like LangChain and Langgraph.

I've also built retrieval-augmented generation (RAG) systems with scalable indexing using Azure AI Search, optimizing fields to reduce storage and achieving up to 60% faster search response times. These real-world projects have given me a good grasp of the latest GenAI techs.

TECHNICAL SKILLS

Frontend Programming	JavaScript, TypeScript
Backend Programming	Python
Frontend Framework	React.js, jQuery
Backend Framework	Django, Fast API, Flask
SQL Database	MySQL, PostgreSQL
NoSQL Database	MongoDB
Data analysis library	NumPy & Pandas
Big data framework	Apache Spark (Pyspark)
Data Platform	Databricks
Machine Learning library	Scikit-learn, Py Torch, Transformers
GEN AI library	Lang chain, Lang graph, Lang Smith, Qdrant, LanceDB, Azure AI Search
Cloud services	AWS, Azure

Employment History

Employer Name: Epam System Private Limited. **Aug 2025 – Present**

(Service based Company)
6th floor, BLOCK-1, DLF Downtown, EPAM Systems India Pvt Ltd, Tharamani, Chennai, Tamil Nadu - 600113

Job Title: Senior Software Engineer

Job/Roles Responsibilities:

- To contribute full-stack enhancements using React frontend and Python/Django backend, integrating GenAI for automated content and user prediction features in a short-term project sprint.
- To support RESTful API development and tweaks for secure frontend-backend data flow, incorporating GenAI for real-time NLP and personalized pharma recommendations.
- Assisted in building responsive React UIs compliant with pharma UX and accessibility standards, leveraging GenAI for dynamic tooltips and tailored dashboards.

Employer Name: Data lens AI Technologies Private Limited. **May 2024 – Jul 2025**

(Service based Company)
7th floor, Arihant Technopolis, 4/293, Rajiv Gandhi salai, Perungudi, Chennai – 600096.

Job Title: Senior Data Engineer

Job/Roles Responsibilities:

- Pipeline Development: Build and optimize ETL/ELT pipelines in Databricks (Spark) to handle large data from various sources efficiently.
- Data Transformation: Use Databricks notebooks and Spark to convert raw data into structured formats for analysis, reports, and ML.
- Cloud Integration: Combine Databricks with AWS to manage data workflows and use cloud storage for big datasets.
- Data Architecture: Create scalable, reliable data systems in Databricks for BI, analytics, and real-time processing.
- Performance Optimization: Monitor and tune pipelines for fast queries and low-cost resource use in Databricks.
- Team Collaboration: Partner with data scientists and analysts to provide clean, usable data for insights and ML models.
- Data Quality: Add checks, cleaning, and governance in Databricks to ensure accurate, secure, and compliant data.

Employer Name: **Cognizant Technology Solutions** **Jan 2024 –Apr 2024**

Cognizant Coimbatore Campus KGISL-SEZ-FO
Special Economic Zone (SEZ), Saravanampatti VIA
Coimbatore North Tal, Keeranatham Village
Coimbatore, Tamil Nadu - 641 035

Job Title: Associate – Projects

Job/Roles Responsibilities:

- Collaborated with cross-functional teams including product managers, QA, and stakeholders to design and implement scalable web-based solutions for internal and customer-facing pharma applications.
- Designed and developed end-to-end software architecture including both client-side React and server-side Python/Django components.
- Built and maintained RESTful APIs to support secure, performant data exchange between front-end and back-end systems.
- Developed clean, responsive, and user-friendly UIs following pharmaceutical industry UX standards and accessibility compliance.
- Managed and optimized relational PostgreSQL databases to ensure data integrity and high availability.

Employer Name: **Aparajitha Software Services Private Limited** **Feb 2020 – Dec 2023**

(Product based Company)
5A Rathinasamy Nadar Road, B. B Kulam, Madurai, Tamil Nadu - 625 002.

Job Title: Python Full Stack Developer

Job/Roles Responsibilities:

- Collaborate with devs and PMs to brainstorm software ideas.
- Design client-side and server-side app structures using React for the frontend and Django for the backend.
- Build appealing front-end interfaces using React.
- Develop and maintain databases and apps using Django on the backend.
- Code APIs to connect React frontend with Django backend and external systems.
- Test for speed and efficiency across React, Django, and cloud deployments.
- Debug, fix, and upgrade software in React frontend and Django backend.
- Implement security and data protection in Django backend and cloud environments.
- Create mobile-responsive features using React.
- Add code comments for easy maintenance in React and Django codebases.
- Document setup, usage, APIs, and architecture (using tools like Swagger/Postman) for React frontend, Django backend, and deployments on Azure & AWS clouds.

Project Details

1. Singapore Fashion Council (Datalens AI Client) – Senior Data Engineer Nov 2024 – Jul 2025

Project details:

Designed and implemented a data pipeline to collect, process, and aggregate data from Edgework, POS, and Shopify platforms, transforming it into a structured Data Lake format using Databricks for daily automation and insights.

Responsibilities:

- Data Collection: Automated the extraction of data from Edgework systems, POS transactions, and Shopify APIs in JSON format.
- ETL Processing: Developed a scalable ETL pipeline using Databricks to clean, normalize, and aggregate raw JSON data.
- Data Transformation: Applied data transformations (e.g., joining datasets, aggregating sales by customer, filtering out irrelevant data) to derive actionable insights.
- Delta Lake Integration: Leveraged Delta Lake format for efficient storage, ACID transactions, and optimized querying of large datasets.
- Automation & Scheduling: Set up and scheduled Databricks Jobs to run daily for consistent, automated data processing.
- Monitoring & Alerts: Implemented monitoring to ensure pipeline reliability and alerted stakeholders in case of job failures.
- Insights & Reporting: Enabled data-driven decision-making by querying the processed data for actionable insights using SQL and visualizing key metrics in Databricks dashboards.

Technology Stack:

- Databricks (Apache Spark, Delta Lake)
- Data Collection: APIs (Shopify, POS, Edgework, Interakt)
- Data Formats: JSON, Delta Lake
- Cloud Storage: AWS S3

2. AI-Powered Resume Review & ATS Optimization for Singapore Fashion Council (Datalens AI Client) – GEN AI Engineer Nov 2024 – Jul 2025

Project Details:

This AI-powered system helps job seekers improve their resumes by ensuring they are optimized for Applicant Tracking Systems (ATS). Using Crew AI, the project analyses resume, compares them to job descriptions, and provides data-driven recommendations for better visibility and higher chances of getting shortlisted.

Responsibilities:

- Designed and implemented an AI-driven system to optimize resumes based on job and company info.
- Built multiple Lang graph agents for resume analysis, job matching, and company research.
- Managed input/output formats (PDF, JSON, Markdown) for smooth data flow.
- Automated generation of tailored resumes and summary reports.
- Tested and integrated all components to ensure accuracy and performance.

Technology Stack:

- Frontend: JavaScript, React.js
- Backend: Python, Fast API, Uvicorn, Crew AI, Chroma, SerperDevtool, ScrapeWebsiteTool, Open AI model, Langgraph.
- Cloud: Azure.

3. Chatbot for Skincare product Requirements
(Unsweetened Beauty – Datalens AI Client) – GEN AI Engineer: Sep 2024 – Nov 2024

Project Details:

Develop and implement a conversational AI chatbot for a skincare brand, offering personalized shopping assistance and answering user queries. The chatbot is designed to enhance the user experience by offering tailored skincare product recommendations, providing skincare advice, and guiding users through their shopping journey. It uses Lang chain and OpenAI models to enable seamless, dynamic, and context-aware interactions.

Responsibilities:

- Chatbot Conversation Design: Build chatbot flows with LangChain for smooth, smart user chats that remember context.
- API Connections: Link in outside APIs to fetch live product info, so the chatbot gives spot-on recommendations.
- Custom Suggestions: Use OpenAI GPT to suggest skincare products tailored to user's skin type, likes, and browsing habits.
- Smart Query Handling: Apply OpenAI GPT to understand questions and reply in natural, friendly ways for skincare tips and product info.
- User Tracking: Monitor how users interact to improve chatbot answers and suggestions over time.
- Testing & Tweaks: Test everything for quick, right responses, then update based on user input and results.
- Launch & Watch: Roll out the chatbot on websites/apps and track its performance with live tools for the best user experience.

Technology Stack:

- Lang chain: Manages conversation flows and external integrations., Langgraph is build to multi agent workflows in the form of nodes & edges.
- OpenAI Models (e.g., GPT): Powers the chatbot's natural language understanding and response generation.
- External APIs: For product data and dynamic recommendations.
- Cloud: Azure.

4. Dhee-AI – Data Platform – Senior Data Engineer (Datalens AI) ***May 2024 – Aug 2024***

Project details:

Dhee is our ingenious platform that offers a comprehensive set of tools and capabilities for streamlining and optimizing data, machine learning operations, and Generative AI applications.

Responsibilities:

- Developed AI-driven solutions for data processing and ML using Databricks to build advanced analytics systems.
- Led data migration and built robust Databricks pipelines for seamless ingestion, transformation, and integration.
- Handled ETL processes to efficiently process, cleanse, and validate large-scale datasets.
- Implemented data quality measures like cleansing, validation, and error-checking for accurate datasets in analytics and AI.
- Managed Azure cloud infrastructure for secure, compliant data operations with governance focus.
- Designed RAG pipelines using OpenAI, Azure AI Search, and LangGraph for natural language queries on structured/unstructured data. Improved retrieval accuracy by 30%, enabled real-time insights without SQL needs, and applied MRL quantization to cut storage costs and latency while preserving semantic accuracy.

Technology Stack:

- Frontend: TypeScript – React.js
- Backend: Python – FastAPI, Databricks SDK Connector
- GENAI Framework: Langchain, Azure AI Search, Qdrant, Langgraph.
- Database: MySQL
- Cloud: Azure

5. Compfie & Compfie Vendor (Programmer Analyst - Aparajitha) **Feb 2020- Dec 2023**

Project details:

ssssIt evaluates performance and functional processes thereby mitigating future risks. either it may be self-tracking or with the support of service provider. From here on, the database will be the storage of all the important data. This will provide the effective security to the system more over the ultimate user can be able to get back with any kind of highly usable reports which he/she would like to see. The system will help to input the specified data in a sequential manner.

Responsibilities:

- Work with development teams and product managers to ideate software solutions.
- Design client-side and server-side architecture. Build the front-end of applications through appealing visual design.
- Develop and manage well-functioning databases and applications.
- Write effective APIs. Software to ensure responsiveness and efficiency. Troubleshoot, debug and upgrade software.

Technology Stack:

- Frontend: Javascript-React.js, jQuery
- Backend: Python-Django, Flask
- Database: MySQL, PostgreSQL
- Cloud: AWS, Azure

CAPSTONE PROJECTS

1. **Airbus Aircraft Customer Support Chat Bot** **GitHub** | **Website**

Project Overview:

The Airbus Aircraft Customer Chat Bot is an intelligent, multi-agent system that provides a seamless, efficient, and personalized experience for users booking flights, hotels, car rentals, and excursions. With its ability to access real-time data via MySQL, interact across multiple service categories, and use AI-powered NLP for context-driven conversations, this bot is a powerful tool for both customers and businesses. By leveraging advanced technologies, the bot helps streamline complex travel arrangements and delivers an exceptional customer experience.

Technology Stack:

- Frontend: JavaScript, React.js
- Backend: Python, Fast API, Uvicorn, Lang graph, Lang chain, Lang chain-google-genAI, Lang smith, MySQL, Faiss DB
- Cloud: AWS Nvidia Ec2, Elastic IP

2. **Amazon Sales Chat Bot** **GitHub** | **Website**

Project Overview:

The amazon sales chat bot is a Self-Reflective Rag, this rag system that provides a seamless, efficient, and personalized experience. This rag reveals a combination of successful transactions and cancellations, offering insight into customer preferences, logistical efficiency, and promotional effectiveness. By analysing these patterns, the business can take steps to optimize fulfilment, reduce cancellations, and tailor marketing efforts to better meet customer needs. Exploring the root causes of issues like cancellations and delays could lead to valuable improvements in both customer service and overall sales strategies.

Technology Stack:

- Frontend: JavaScript, React.js
- Backend: Python, Fast API, Uvicorn, Lang graph, Lang chain, Lang chain-google-genAI, Llama Index, Qdrant, Tavily Search
    - **Cloud:** AWS Nvidia Ec2, Elastic IP

3. **You Tube Video Summarization** **GitHub** | **Website**

Project Overview:

The You tube Video Summarization focuses on creating a multimodal summary of YouTube videos by extracting and combining both audio and visual content. It begins by extracting the video and audio, then transcribes the spoken content into text. Key images or frames from the video are also extracted to capture visual elements. The transcribed text and images are indexed and stored for quick retrieval. Finally, a multimodal model processes both the text and images to generate a comprehensive summary, providing a richer and more detailed overview of the video.

Technology Stack:

- **Frontend:** JavaScript, Recat.js
- **Backend:** Python, Fast API, Uvicorn, Llama Index, Lance DB, Google Model, Transformers, OpenAI- Whisper, Clip Embeddings, Torch, Torch Vision
- **Cloud:** AWS Nvidia Ec2, Elastic IP

4. **What's App AI Agents** **GitHub | Website**

Project Overview:

Integrating Lang graph, Groq, Eleven Labs, and Together.ai to create AI agents for WhatsApp enables a highly interactive, multimodal user experience. By allowing text-to-image, image-to-text, text-to-audio, and audio-to-text capabilities, businesses can offer richer, more personalized interactions with users. Whether for customer service, content creation, or education, this technology can revolutionize the way users engage with brands and services on WhatsApp.

Technology Stack:

- **Frontend:** JavaScript, React.js
- **Backend:** Python, Fast API, Uvicorn, Lang graph, Qdrant-client, Llama model, Llama Vision model, OpenAI- Whisper, Together.ai, IElevenLabs
- **Cloud:** AWS Nvidia Ec2, Elastic IP

5. **Singapore resale flat price** **GitHub | Dags Hub | ML Flow**

Project Overview:

The DS-Singapore-Flat-Resale-Price project is designed to predict the resale price of flats in Singapore using a regression model. The project employs a combination of machine learning, data visualization, and model management tools to predict flat prices based on various features such as flat size, location, age, and amenities. The main goal is to predict flat resale prices accurately and create an interactive environment where users can input different flat characteristics and get predictions in real-time.

Technology Stack:

- **Backend & UI:** Python, Streamlit for UI, Scikit-learn, XG Boost, Mlflow, Dagshub, plotly

Courses

Mastering Generative AI with OpenAI, Lang chain, and Lama Index v2 **May 2024 – Aug 2024**
Ineuron.ai Institute - Bangalore

Data Science Professional with IIT-M Pravartak Certification **May 2024 – Dec 2024**
Guvi Institute - Chennai

EDUCATION

Bachelor of Engineering (Mechanical Engineering)

Anna University College of Engineering - Dindigul Campus **2015 - 2019**